



PROPYLENE GAS (PG)

The Uhde STAR Process – A Case Study

Steam Active Reforming (STAR) Process

OXYDEHYDROGENATION OF LIGHT PARAFFINS TO OLEFINS

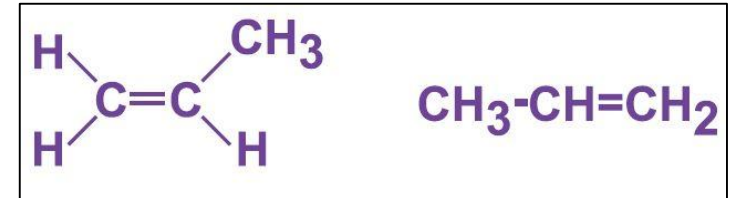
Dr. Sahar El-Marsafy – Professor – Chemical Engineering Department



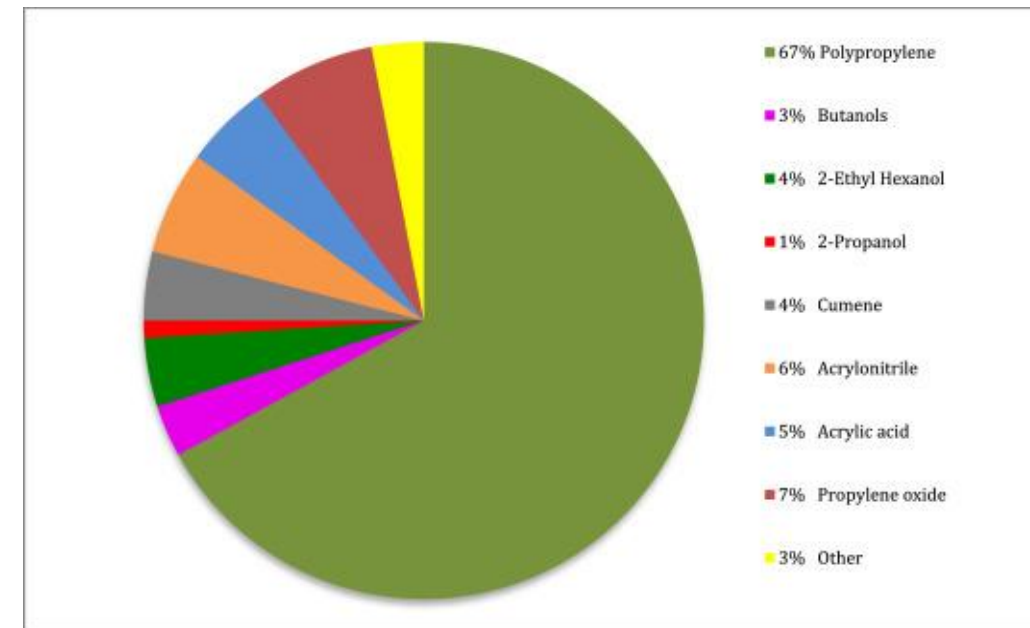
About Propylene Gas (PG)

At room temperature and pressure, propylene is a gas. It is colorless, odorless (an odor similar to garlic is sometimes added to propylene to make it easily detectable), and highly flammable

Systematic name	Propene
Molecular formula	C_3H_6
Molar mass	42.08 g/mol
Appearance	colorless, odorless gas
Density and phase	0.52 g/cm ³ , liquid, 20 °C
Solubility in water	0.61 g/m ³ (25 °C)
In ethanol, ether	Soluble
Melting point	- 185.2 °C (88.0 K)
Boiling point	- 47.6 °C (225.5 K)



World PG Supply Demand



About Propylene Gas (PG)

- Propylene (major chemical intermediate) is the **second most important** starting product in the petrochemical industry after ethylene.
- Propylene is produced primarily as a by-product of petroleum refining and of ethylene production by steam cracking of hydrocarbon feedstock (**Olefins production plant**).
- The largest volume of PG is produced from NG or naphtha in steam cracking processes. In both of these processes, PG is a **byproduct of ethylene**. However, it can also be manufactured through on-purpose processes in which the main product is PG. The main processes for on-purpose PG are Propane Dehydrogenation (PDH), Metathesis, Methanol-to-Olefins (MTO) and Olefins Catalytic (OCT) processes.
- The most important derivatives of chemical and polymer grade propylene are polypropylene, acrylonitrile, propylene oxide, isopropanol and cumene (isopropyle benzene).
- Use of polypropylene in plastics (injection molding) and fibers (carpets) accounts for over one-third of PG consumption. It is also used in the production of synthetic rubber and as a propellant or component in aerosols.

Egyptian Propylene & Polypropylene Company

- Egyptian Propylene & Polypropylene Company (EPPC), 50:50 joint venture between Egypt Oriental Petrochemicals Company and the state-owned Egyptian Petrochemical Holding Company (ECHEM), under the lead of Oriental Weavers Group, has commissioned Uhde to build a turn-key petrochemical complex in Port Said, some 170 kilometers north-east of Cairo.
- The complex consists of a propylene plant and a polypropylene plant with respective annual production capacities of 350,000 tones as well as all utilities and off sites, including an air fractionation and refrigerating unit, and the required storage tanks.
- The contract signed in Cairo in 2006 includes the license, basic and detail engineering, supply of equipment, construction, training of the operating personnel and commissioning. EPPC is investing some US\$ 680 million in this new petrochemical complex. Completion was scheduled for late 2009. Propane from Egyptian natural gas deposits is used as the feedstock.

Specifications - Egyptian Propylene & Polypropylene Company (EPPC), Egypt

Plant owner	Egyptian Propylene & Polypropylene Company (EPPC)
Project location	Port Said, Egypt
Output	Propylene and 350,000 tpa polypropylene (PP)
Investment	\$ 680 MM
Startup	2009
Construction contract	Uhde
Propylene process technology	Uhde (STAR process)
Polypropylene process technology	Basell

The Uhde STAR Process

OXYDEHYDROGENATION OF LIGHT PARAFFIN TO OLEFINS

- **STAR**, the acronym for STeam Active Reforming (STAR), is a commercially established technology for the dehydrogenation of light paraffins, such as propane or butane, to produce olefins.
- The **STAR** process was the first to apply the oxy-dehydrogenation principle for propylene production, significantly enhancing its performance. In this technology, oxygen is added to the system to react with hydrogen, forming water.
- This reaction lowers the hydrogen partial pressure, shifting the equilibrium to higher conversion of propane into propylene while providing the required heat of reaction.

Dehydrogenation – Basic Principles

1. Conventional Dehydrogenation

- Dehydrogenation is an endothermic equilibrium reaction.
- The conversion increases with decreasing pressure and increasing temperature.
- As shown below for propane and butane, respectively, the major reaction is the conversion of paraffin to olefin.
- Propane dehydrogenation (PDH);
- Butane dehydrogenation (BDH):
- Lower hydrocarbons (i.e. lower in carbon number than that of the feedstock) are also formed.
- Minor reactions that occur are cracking, which is primarily thermal and results in the formation of small amounts of coke.

Propane dehydrogenation (PDH):



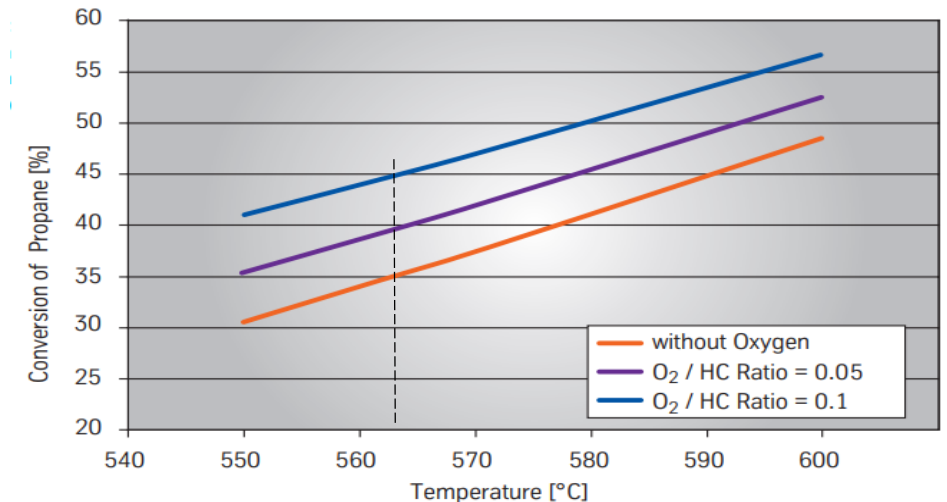
Butane dehydrogenation (BDH):



Dehydrogenation – Basic Principles

2. Oxy-Dehydrogenation

- Obviously conversion is limited by the thermodynamic equilibrium for a given pressure and temperature.
- As conversion approaches equilibrium, reaction velocity decreases and catalyst activity is not efficiently utilized.
- However, if oxygen is introduced into the system it will form H_2O with part of the hydrogen from dehydrogenation, which in turn will shift the equilibrium of the dehydrogenation reaction towards increased conversion of propane.
- The chart shows the influence of oxygen addition which shifts the equilibrium towards increased conversion of propane to propylene.
- Catalytic formation of H_2O is exothermic and hence provides the heat of reaction for further endothermic conversion of paraffin to olefin.

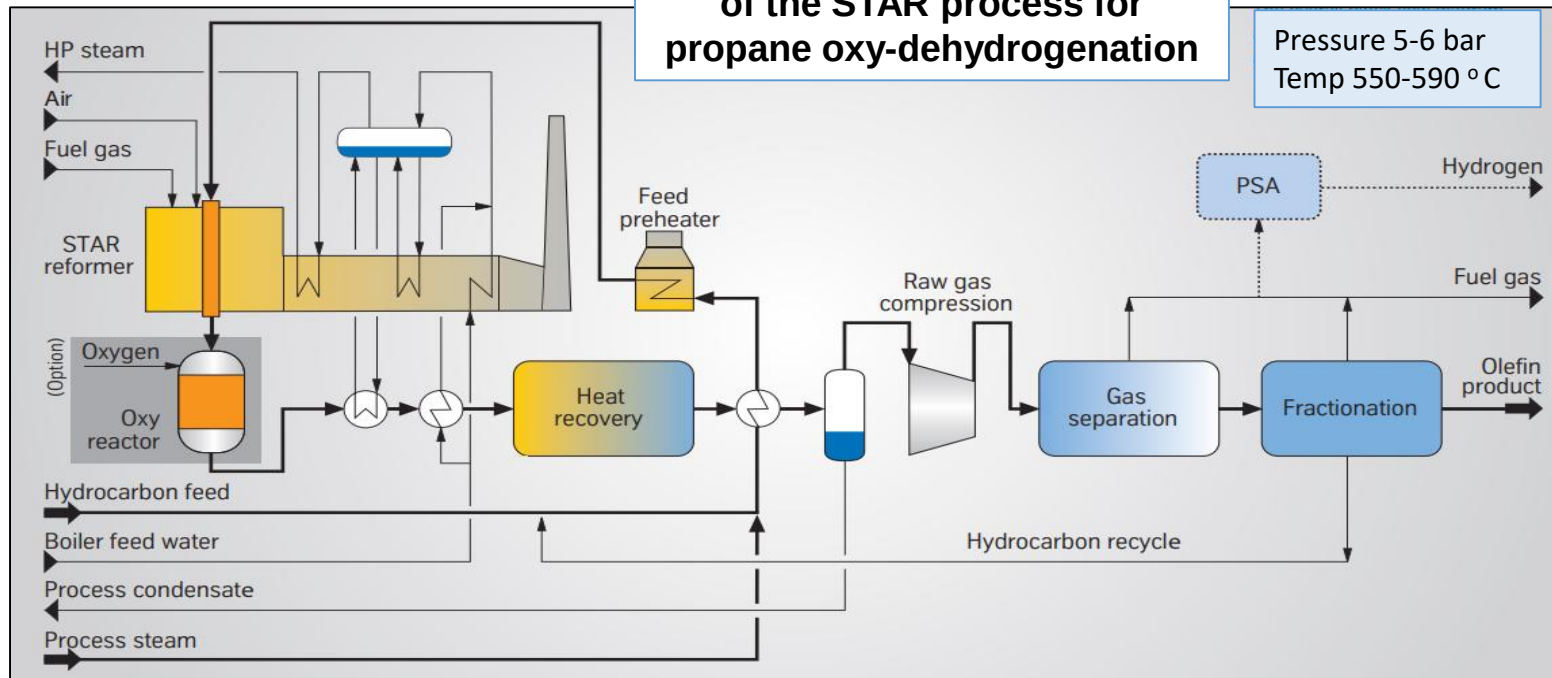


Thermodynamic equilibrium data
(Partial pressure 1 bar, molar oxygen to propane ratios)

STAR process consists of four major steps:

-
- The diagram illustrates a chemical process flow. It begins with 'Fresh Feed' entering a junction point. At this junction, 'Process Steam' (indicated by a downward arrow) and 'Oxygen/Air' (indicated by a downward arrow) are added. The mixture then flows into a 'Reaction' unit. The output of the 'Reaction' unit goes to 'Gas Compression', which then feeds into 'CO2 & Gas Separation'. From 'CO2 & Gas Separation', a stream goes to 'Fractionation'. The 'Fractionation' unit has two outputs: one goes to 'Fuel' (indicated by an upward arrow) and another goes to 'Product' (indicated by a rightward arrow). A 'Recycle' stream (indicated by a downward arrow) returns from the 'Fractionation' unit to the initial junction point before the 'Reaction' unit.

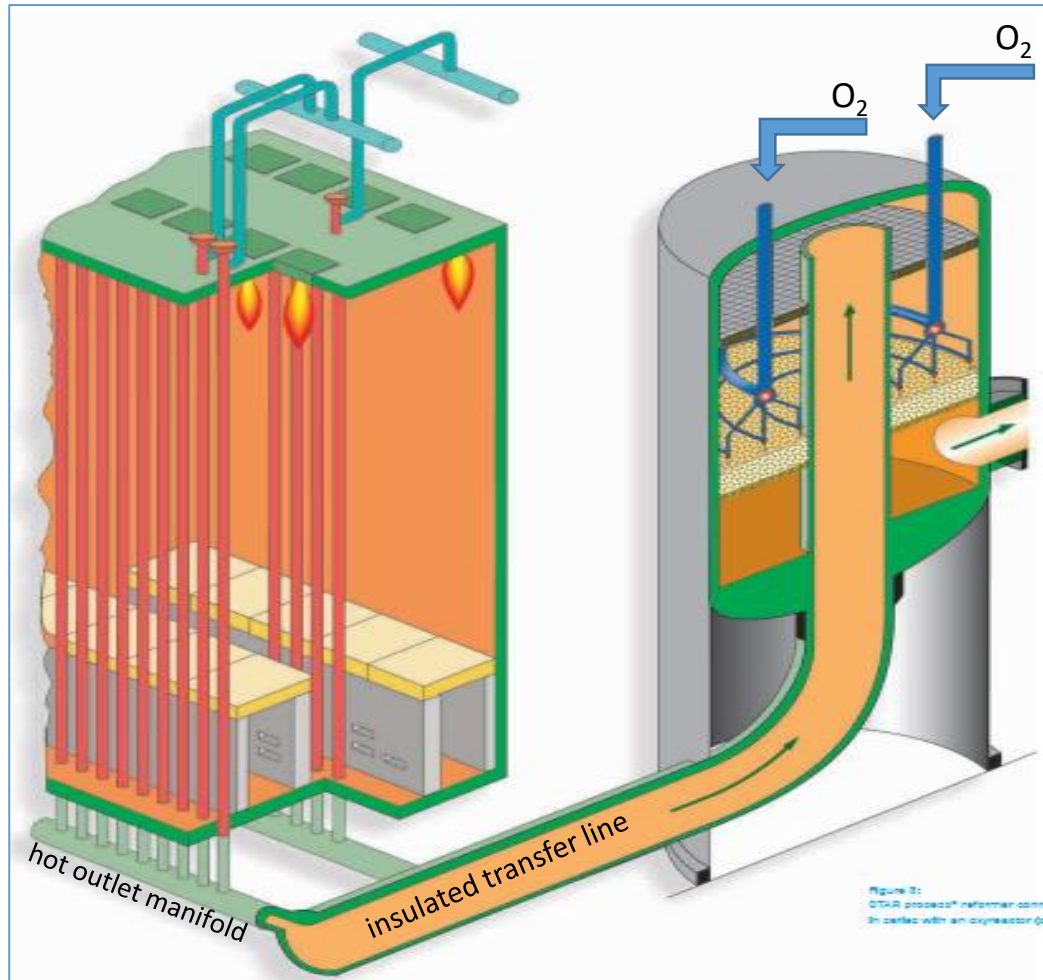
The overall block flow scheme of the STAR process for propane oxy-dehydrogenation



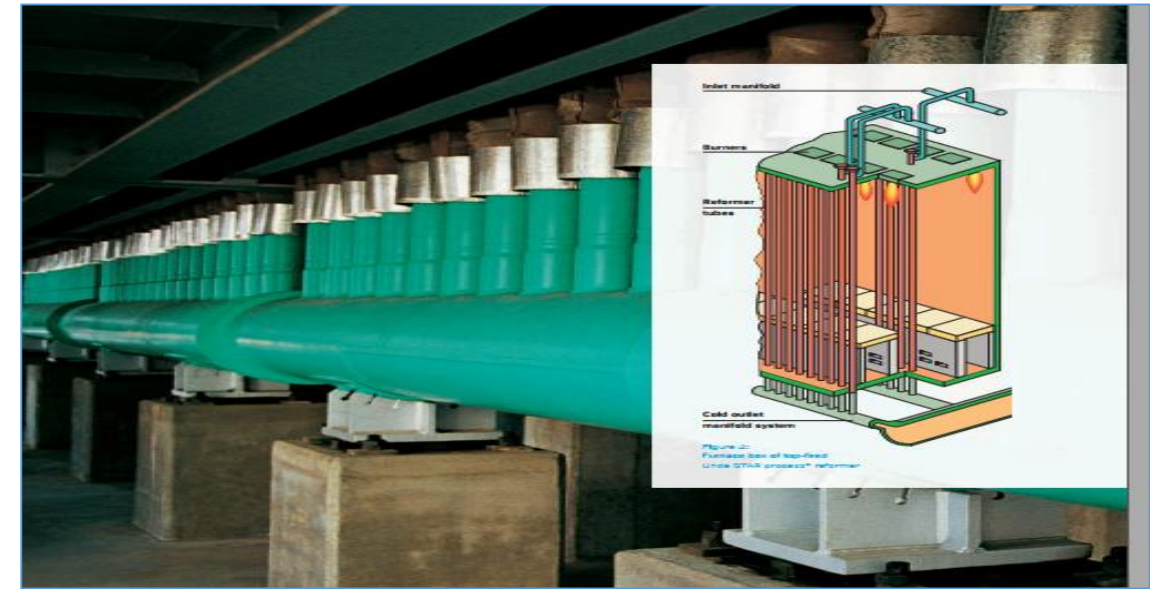
- HC feed is vaporized & superheated
- Boiler feed-water is heated
- Product stream is cooled
- HP steam is generated
- Fractionation column re-boilers are directly heated

Top fired steam reformer of the tubular fixed bed type

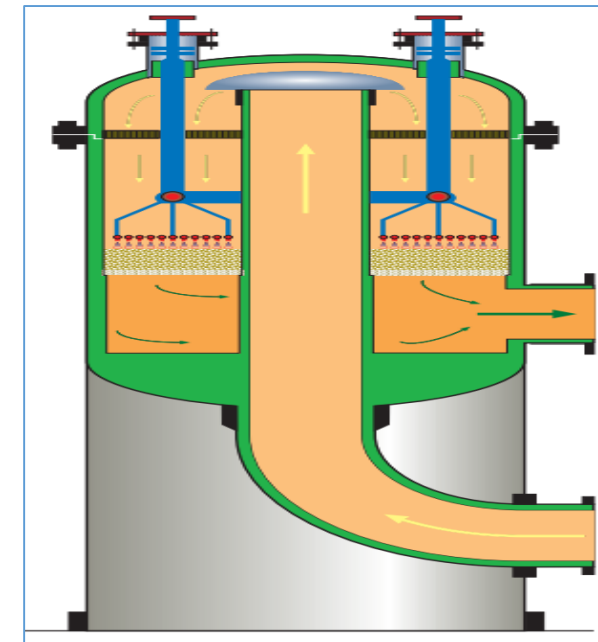
STAR Process Reformer connected in series with the oxy-reactor



STAR Process Reformer



zinc and calcium aluminate catalyst
550-590°C
5-6 bar



STAR
Process
Oxy-
reactor

Uhde STAR Technology- Reaction Section

- The previously mentioned advantages of oxy-dehydrogenation are utilized in the STAR process® which connects conventional dehydrogenation (i.e. the STAR reformer) in series with an oxy-dehydrogenation step (i.e. oxy-reactor).

Reaction Section

- The reaction section comprises an externally heated reactor (STAR Process Reformer) including a highly efficient heat recovery system for feed preheating and steam generation. Optionally, the reaction section is equipped with an oxy-reactor to increase the conversion rates.

STAR Process Reformer

It is a commercially established top fired steam reformer of the tubular fixed bed type. Its main features are:

- Top firing for optimum uniformity of tube skin temperature profile.
- Internally insulated transfer line made from carbon steel and located externally under the reformer bottom.
- Each tube row is connected to separate hot outlet manifold. Each hot outlet manifold line is connected to a main cold transfer line.

Uhde STAR Technology- Operation Cycle

- **The STAR catalyst®**
 - It is based on a zinc and calcium aluminate support that, impregnated with various metals, demonstrates excellent dehydrogenation properties with very high selectivity at near equilibrium conversion. Due to its basic nature it is also extremely stable in the presence of steam and oxygen at high temperatures.
- **Operation Cycle**
 - Dehydrogenation is a strongly endothermic equilibrium reaction, in which propane is converted to propylene. Meanwhile, lower hydrocarbons like ethane, ethylene and methane are also formed. Minor reactions that occur are cracking, which is primary thermal and results in the formation of small amounts of coke deposited on the catalyst. **This requires regular regeneration of the catalyst to burn off coke**.
 - **Steam present in the system converts most of the deposited coke to carbon dioxide**. This leaves very little coke to be burnt off during the oxidative regeneration, thus allowing longer operation cycles and making the regeneration quick and simple.
 - **The cycle length is seven hours of normal operation followed by a regeneration period of one hour**. Therefore only 14.7% additional reactor capacity is needed to account for regeneration requirements, which is least of all commercial technologies.

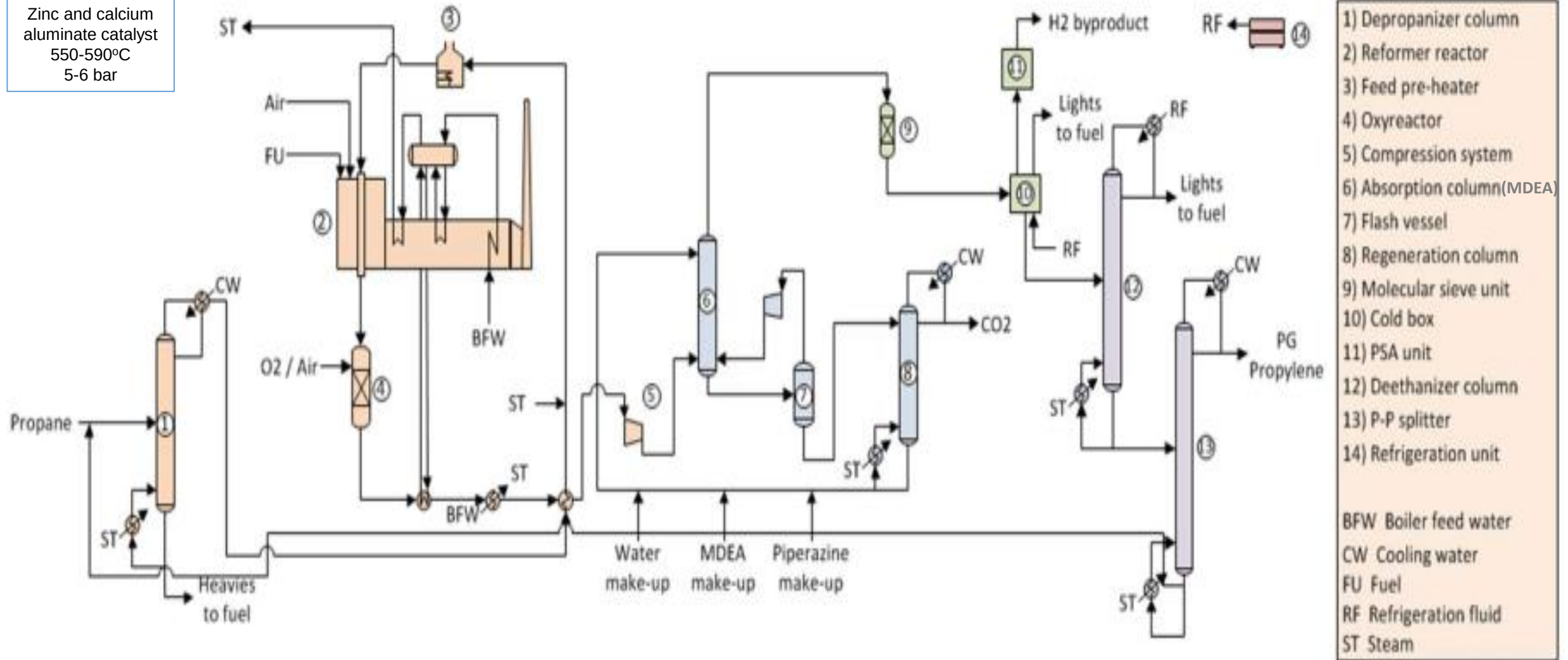
The Uhde STAR Process: Overview

- Beside the four main process steps, some other process units are used to provide and supply the different required utilities or handle processes effluents such as:
 - Ammonia refrigeration system
 - Condensate treatment
 - Waste water system
- Feedstock to the propylene de-hydrogenation (PDH) plant is fresh propane from OSBL (Out Side Battery Limit) and added to recycled stream.
- The consumption figures for propane oxy-dehydrogenation using STAR process (per metric ton of polymer grade propylene produced) is as follows:

Feed (Kg paraffin/t):	1,188
Fuel (GJ/t):	2.1
HP Steam (t/t):	2.2
Electric Energy (KWh/t):	99
Cooling Water (m ³ /t):	155
Nitrogen (Nm ³ /t):	4

Uhde STAR Process Flow Diagram

Zinc and calcium
aluminate catalyst
550-590°C
5-6 bar



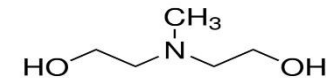
Uhde STAR Technology- Process Description

Reaction

1. Propane is fed into a depropanizer column for heavy impurities removal.
2. The depropanizer distillate is vaporized, mixed with steam, and pre-heated before entering the reformer, which is an externally fired tubular reactor where most of the propane conversion occurs.
3. Next, the reformer effluent is fed to the oxy-reactor, where hydrogen is selectively reacted with oxygen forming water in an exothermic reaction.
4. The process gas is cooled in a series of heat exchangers to recover process heat. Then, it is compressed and sent to the next area.

CO₂ Separation

5. The compressed gas is fed to an absorption column, where CO₂ is removed in the bottoms by washing with an aqueous piperazine-activated methyldiethanolamine (MDEA) solution.



6. The bottoms stream is depressurized, liberating any dissolved gases that are recycled to the absorption column.
7. The CO₂-rich liquid solution from the flash vessel is fed to the solvent regeneration column, which separates CO₂ in the overheads from the MDEA solution in the bottoms, which is recycled to the absorption column.

Uhde STAR Technology- Process Description

Gas Separation

8. The overhead stream of the absorption column is dried in a molecular sieve and fed to a low-temperature separation unit (cold box) the objective of which is to separate hydrogen and light byproducts (generated in the reaction step) from the main product.



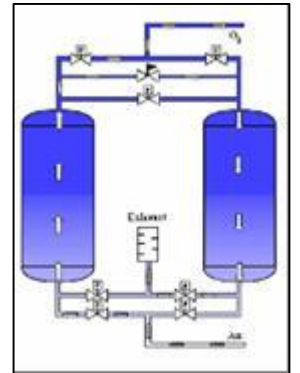
Molecular Sieve Spherical beads of aluminosilicate



9. In this unit, a hydrogen-rich stream is also obtained, which is purified in a Pressure Swing Adsorber (PSA) unit to obtain high-purity hydrogen byproduct.

Pressure swing adsorption (PSA) is a technique used to separate a specific gas from a mixture of gases under pressure. Selective adsorbent materials are used as trapping material, preferentially adsorbing the target gas at high pressure. The process then swings to low pressure to desorb the adsorbed gas.

Pressure swing adsorber (PSA)



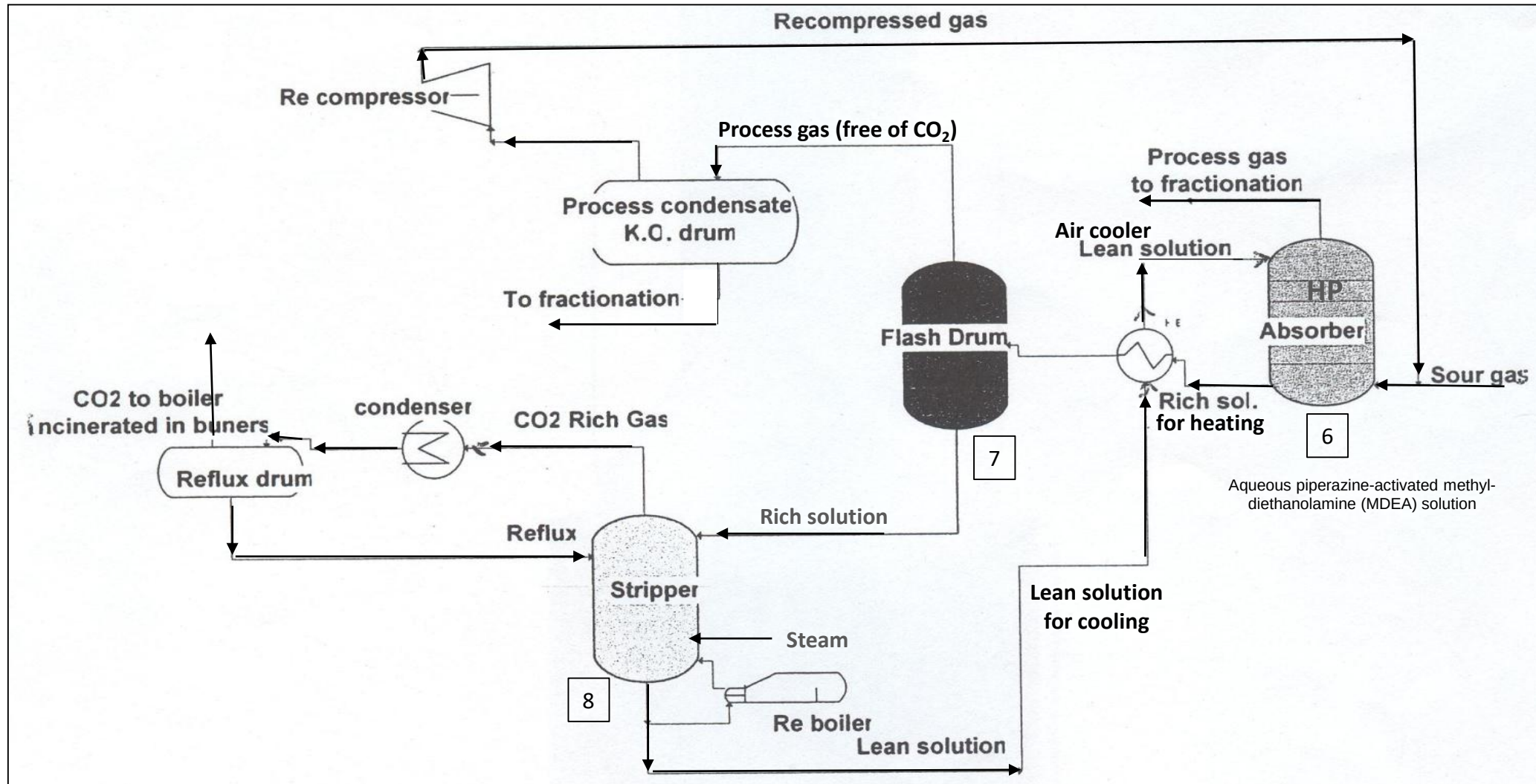
Fractionation

10. The liquid fraction from the cold box is sent to the fractionation area for product recovery.
11. The fractionation facilities consist of a de-ethanizer column for removal of lights and a propylene-propane (P-P) splitter, where polymer-grade propylene is obtained.
12. Unreacted propane from the P-P splitter bottoms is recycled to the reaction area.

CO₂ Separation

- The compressed sour gas is fed to an absorption column, where CO₂ is removed in the bottoms by washing with an aqueous piperazine-activated methyl-diethanolamine (MDEA) solution, while as the process gas (almost free of CO₂) is sent to the fractionation section (after being dried and deep cooled in the cold box).
- The rich solution leaving the bottom of the absorber is flashed in the HP flash drum. Flash gas from the flash drum includes small amounts of co-absorbed hydrocarbons and thus it is routed to the process condensate drum, recompressed and reprocessed in the absorber to minimize losses of propylene.
- The rich solution leaving the flash drum is introduced into a stripper for boiling-off the absorbed CO₂ using steam. Carbon dioxide and associated water vapor leaving the stripper are condensed in a stripper condenser and sent to a stripper reflux drum. The condensed liquid is returned as stripper reflux, while the gas containing CO₂ is routed to an auxiliary boiler unit, where it is combined with the combustion air to be incinerated in the burners.
- The lean solvent obtained as bottom product of the stripper is cooled against the rich solvent leaving the absorber. Then further cooled using air cooler and recycled back to the absorber. A slip stream of the lean solvent is filtered using cartridge filters, then charcoal filters to prevent the enrichment of heavier components in the solvent loop.

CO₂ Separation Package – Flow Diagram



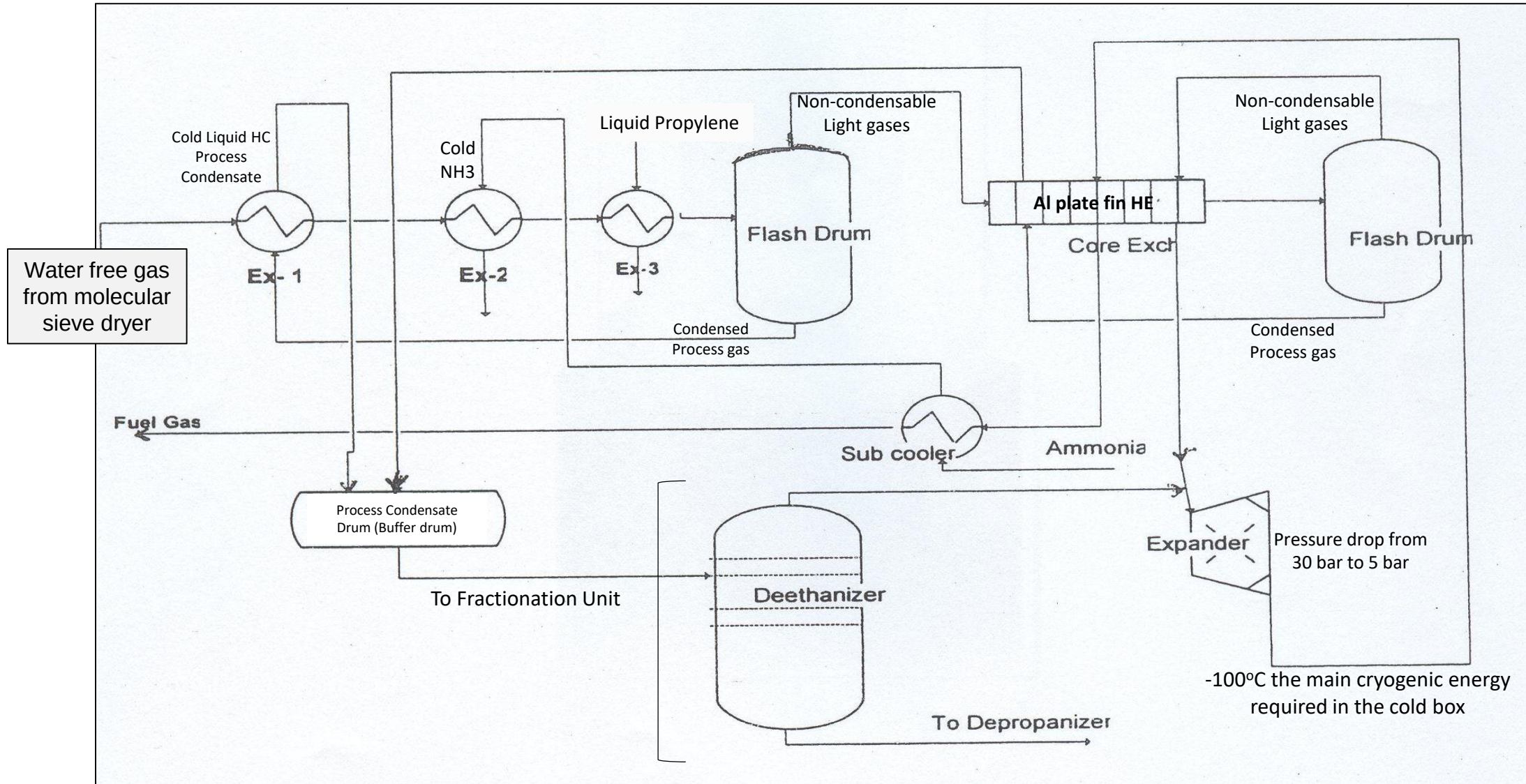
Uhde STAR Cold Box

- The purpose of the cold box is to separate incondensable process gas components like hydrogen, nitrogen, and methane from the propane and propylene containing phase by partial condensation. 70% of the methane and 100% of the hydrogen and nitrogen are separated in the cold box. The hydrocarbon phase containing C2 and C3 will be condensed while hydrogen, nitrogen, and methane remain in the gas phase.
- Water free gas from the dryer is further partially condensed by cold liquid hydrocarbon, refrigerant ammonia (-10°C) and refrigerant propylene in the dried gas chillers I, II, and III.
- Gas phase and liquid phase (Propane, PG & C₂ gases which dissolve in the liquid phase) are separated in the dried gas flash drum. The liquid phase is pumped to the de-ethanizer feed drum (buffer drum to collect process gas to deliver it to the fractionation unit **with constant flow rate and composition**). The gas phase is sent to the core heat exchanger (aluminum fin plate HE) of the cold box for further refrigeration.
- The aluminum fin plate HE has the advantage of low temperature approach and it withstands very low temperature with no embrittlement. The temperature in the core HE reaches -100°C.

Uhde STAR Cold Box

- The core exchanger is a multi-stream-capable heat exchanger in which heat exchange of deep cold effluents with cold box feed gas, which has to be chilled, takes place. The cold process gas from the chillers enters the core HE for further refrigeration and thus partial condensation.
- Due to the refrigeration of the process gas, more hydrocarbons will condense out of the process gas. The liquid phase is separated from the gas phase in a flash drum and is pumped to the de-ethanizer feed drum after exchange in the core exchanger and dried gas chiller.
- The remaining gas phase, which is obtained deep cold, is also sent back to the core exchanger where it serves as refrigerant media.
- By heat exchange with cold box feed gas it (the remaining gas phase) is warmed up and together with the overheads products gas from de-ethanizer rectifier they are sent to the expander.
- Due to the polytropic expansion from approximately 30.6 kg/cm² (abs) to 5.1 kg/cm² (abs) it cools down to temperature below -100°C and supplies the main part of the cryogenic energy required in the core exchanger.
- Refrigerant from the core heat exchanger is sent via refrigerant sub-cooler (where liquid ammonia is cooled to -10°C) to the fuel gas header. Mechanical energy is available at the coupling of the expander and is used for generation of electrical power, which is charged into the electrical power grid.

Uhde STAR Cold Box (10) – Flow Diagram



Uhde STAR Fractionation Unit

The distillation section consists of three column systems:

1. Deethanizer Stripper / Deethanizer Rectifier
2. C3 Splitter
3. Depropanizer

- C3 splitter separates propane from propylene, heavier are removed from propane feed stock in the Depropanizer.

The De-ethanizer Unit:

- It consists of two columns, stripper and rectifier (WHY?). In the Deethanizer stripper and Deethanizer rectifier column, ethane, ethylene and other light components (30% methane not removed in the cold box) are removed from the hydrocarbon liquid phase which is obtained after partial condensation in the cold box.
- The feed to the stripper, coming from the treated gas flash drums as well as from dried gas flash drum and the upstream cold box, is collected in the deethanizer feed drum, which is used as a buffer to maintain a constant flow and composition to the following distillation section over the cycles of normal and regeneration modes of reaction section.

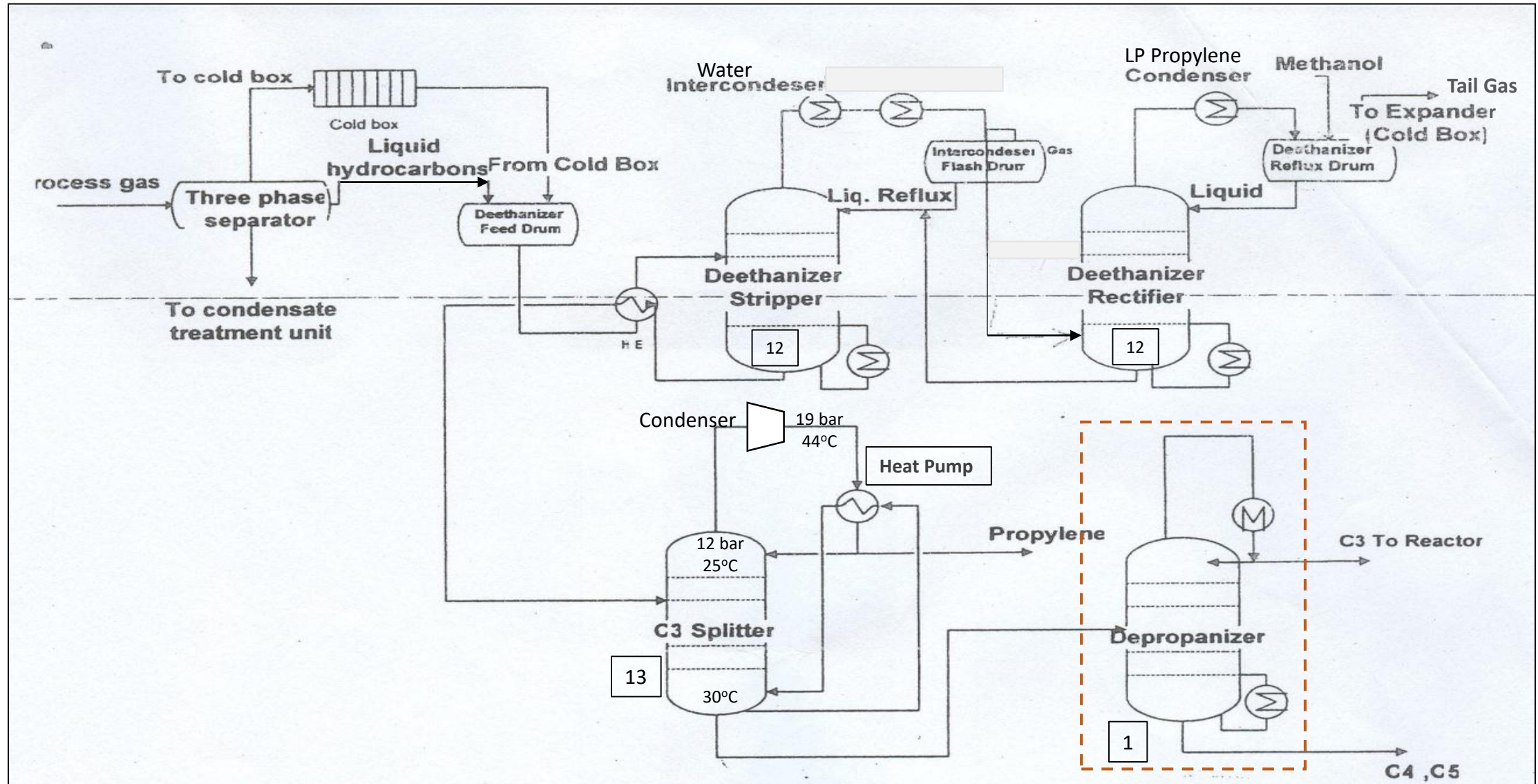
Uhde STAR Fractionation Unit

- The stripper feed is pumped from the deethanizer feed drum to be preheated (in the deethanizer feed / deethanizer bottoms heat exchanger) with the hot deethanizer bottom product before entering the column.
- The bottom stream of the deethanizer stripper (mixture of C3 compounds, i.e. propylene and propane) is charged via deethanizer feed/bottoms exchanger to the C3 splitter which separates propane and heavier components from product propylene.
- An inter-condenser cooled by water is introduced at the top of the deethanizer stripper.
- Vapors leaving at the top of the stripper are partially condensed with cooling water in the deethanizer inter-condenser, and the two phase stream leaving the condenser is fed to the inter-condenser flash drum, where the liquid flash stream is sent to the top of the deethanizer stripper as reflux, and the gaseous flash stream is introduced into the bottom of the deethanizer rectifier.
- The deethanizer-rectifier condenser operates at the lowest temperature level using low pressure propylene as refrigerant.

Uhde STAR Fractionation Unit

- Bottom liquid from rectifier is sent together with inter-condenser reflux liquid to deethanizer stripper as reflux.
- The lights-containing overhead vapors from the top of the deethanizer rectifier are partially condensed in the deethanizer rectifier condenser (using low pressure propylene), where the liquid portion is accumulated in the deethanizer reflux drum and returned to the top of the rectifier column as reflux.
- Non condensable gas is sent to the cold box where it is directed to the expander where its pressure drops from 30 bar to 5 bar and its temperature drops to -100°C , thus providing the main cryogenic energy required in the cold box and after cryogenic exchange in the core exchanger and refrigerant sub-cooler it is admixed with the gas phase and then sent to the fuel gas header.
- To prevent formation of hydrates in the rectifier, methanol is continuously pumped to the deethanizer reflux drum.

Uhde STAR Fractionation Unit– Flow Diagram



Ammonia Refrigeration System

- Low level heat from the process gas is utilized in the NH_3 absorption refrigeration system to provide refrigeration duty in the process. It is used to fractionate pure ammonia from an $\text{NH}_3\text{-H}_2\text{O}$ mixture. The liquid NH_3 is used as refrigerant to reduce the requirements for propylene refrigerant supply.
- Liquid NH_3 is produced at a pressure of 3.6 kg/cm^2 abs., at a temp. of approx. -10°C and at a pressure of 6.2 kg/cm^2 abs., corresponding to $+10^\circ\text{C}$ (HP. NH_3).
- **Advantages of NH_3 Refrigeration System:**
 - Flexible operation and quick adjustment to varying loads.
 - Reliable design with pumps as only rotating equipment and thus low maintenance.
 - Utilization of waste energy from the process flow.

STAR Process Major Features

- A steady continuous feed provided by the de-ethanizer feed buffer drum to the fractionation at a constant composition is important to ensure the product quality.
- The cold box virtually removes all the hydrogen and nitrogen from the reactor product. About 70% of methane and minor amounts of ethane are also removed.
- Olefin losses are very low. Olefin recovery in the product is 99.5% of the olefin produced in the reactor.
- No gas phase is admitted to the fractionation. The entire fractionation feed is liquid, which is collected in an intermediate storage vessel (de-ethanizer feed buffer drum), from where it is continuously fed to the distillation columns. Hence operation of this section is not influenced by the load variations (between the normal and regeneration mode) in the front end.
- Light ends are removed as tail gas in the fractionation. The removal of hydrogen, nitrogen and a substantial amount of methane in the cold box reduces the amount of these components in the light ends. This increases the dew point of the tail gas for a given pressure. As a result, the required temperature level of the refrigerant is also relatively higher, which in turn reduces compression costs.

Heat Recovery (V. I.)

Heat Recovery in the Reaction Zone

- **Heat from the process gas is efficiently recovered and utilized for:**
 - Feed vaporization and superheating
 - Direct heating of the fractionation columns re-boilers
 - Generation of refrigerant
 - Generation of high pressure steam
 - Preheating of boiler feed water

- Give examples for energy conservation and integration in each area of propylene production process.
 - Reaction area
 - CO₂ Removal Unit
 - Cold Box
 - Fractionation Unit